Your grade: 100%

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Next item

1. True or false? 1 / 1 point

The derivative of a function at a point x represents the instantaneous rate of change at x.

- (•) True
- ( ) False

Nice work

The average rate of change corresponds to the slope of a secant line. The instantaneous rate of change corresponds to the slope of a tangent line, which is the derivative.

2. Mary is competing in the 100-meter dash. Suppose that her position is described by the position function $p(t)$, where t is the time in seconds and $p(t)$ is her position in meters. If she finishes the race (100 meters) in 15 seconds, and her instantaneous velocity at $t = 1.5$ seconds is $6.2\frac{\text{meters}}{\text{seconds}}$, what is the slope of the line tangent to the graph of $p(t)$ at $t = 1.5$? 1 / 1 point

- (•) 6.2
- ( ) There is not enough information
- ( ) 4.1
- ( ) 6.7

Nice work

The slope of the line tangent to the graph of $p(t)$ at $t = 1.5$ is equal to Mary's instantaneous velocity at $t = 1.5$, or $6.2\frac{\text{meters}}{\text{second}}$.

3. Find the equation of the secant line to the curve $f(x) = 2x^2 - 3$ on the interval $[1, 1.2]$ at $x = 1$. 1 / 1 point

- (•) $y = 4.4x - 5.4$
- ( ) $y = 5.6x + 6.6$
- ( ) $y = 5.6x - 6.6$
- ( ) $y = 4.4x + 5.4$

Nice work

To determine a line, you need a point and the slope. Start by finding the slope.

$$m = \frac{f(1.2) - f(1)}{1.2 - 1} = \frac{(-0.12) - (-1)}{0.2} = 4.4$$

Now $(1, f(1)) f(1) = -1$, so the point is at $(1, -1)$.

Now use the point-slope form of a line where $x_1 = 1$ and $y_1 = -1$.

$y - y_1 = m(x - x_1)$

$y + 1 = 4.4(x - 1) y = 4.4x - 5.4$

4. If the tangent line to $y = f(x)$ at the point $(4, 3)$ passes through the point $(0, 2)$, find $f'(4)$. 1 / 1 point

If your answer is not a whole number, enter it as a decimal.

0.25

Nice work

$f'(4)$ is the slope of the tangent line to f at $(4, 3)$. The points $(4, 3)$ and $(0, 2)$ lie on this line, so we can solve for the slope:

$$f'(4) = \frac{3 - 2}{4 - 0} = \frac{1}{4} = 0.25$$

5. The following limit represents the derivative of a function f at a number a. 1 / 1 point

$$\lim_{x \to 1/4} \frac{\frac{1}{x} - 4}{x - \frac{1}{4}}$$

Find f and a.

- (•) $f(x) = \frac{1}{x}$, $a = \frac{1}{4}$
- ( ) $f(x) = \frac{1}{x}$, $a = 4$
- ( ) $f(x) = \frac{1}{x - \frac{1}{4}}$, $a = 0$
- ( ) $f(x) = \frac{1}{x - \frac{1}{4}}$, $a = 4$

Nice work

Compare the limit to the definition of the derivative: $f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$.

Let $h = x - \frac{1}{4}$. Then as $x \to \frac{1}{4}, h \to 0$ and we can write

$$\lim_{x \to 1/4} \frac{\frac{1}{x} - 4}{x - \frac{1}{4}} = \lim_{h \to 0} \frac{\frac{1}{h + \frac{1}{4}} - 4}{h} = \lim_{h \to 0} \frac{\frac{4}{h + 1} - \frac{1}{4}}{h}$$

Thus, $f(x) = \frac{1}{x}$ and $a = \frac{1}{4}$

6. Let $f(x) = x^2 \sin \frac{1}{x}$ if $x \neq 0$ and $f(x) = 0$ if $x = 0$. 1 / 1 point

Determine whether $f'(0)$ exists.

- (•) $f'(0)$ exists.
- ( ) $f'(0)$ does not exist.

Nice work

Using the limit definition of the derivative,

$$f'(0) = \lim_{h \to 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \to 0} \frac{h^2 \sin \frac{1}{h} - 0}{h}$$

$$= \lim_{h \to 0} h \sin \frac{1}{h}$$

We can find this limit using the Squeeze Theorem.

$-1 \leq \sin \frac{1}{h} \leq 1,$

thus when $h \geq 0, -h \leq h \sin \frac{1}{h} \leq h$

and when $h < 0, h \leq h \sin \frac{1}{h} \leq -h$

We know that $\lim_{h \to 0} h = \lim_{h \to 0} -h = 0.$

Thus we can use the Squeeze Theorem on the lefthand and righthand limits at 0 of $h \sin \frac{1}{h}$ to obtain

$$\lim_{h \to 0} h \sin \frac{1}{h} = \lim_{h \to 0^+} h \sin \frac{1}{h} = \lim_{h \to 0^-} h \sin \frac{1}{h} = 0$$

and thus $f'(0) = 0$.

7. Using the definition of the derivative, find the derivative of the function $g(x) = -4x + 2$. 1 / 1 point

-4

-4

Nice work

Use the limit definition of the derivative.

$$g'(x) = \lim_{h \to 0} \frac{g(x + h) - g(x)}{h}$$
$$= \lim_{h \to 0} \frac{[-4(x + h) + 2] - [-4x + 2]}{h}$$
$$= \lim_{h \to 0} \frac{-4x + 4x + 2 - 2 - 4h}{h}$$
$$= \lim_{h \to 0} \frac{-4h}{h}$$
$$= \lim_{h \to 0} (-4)$$
$$= -4$$

8. Let $f(x) = \begin{cases} x^2, & \text{if } x \leq 4 \\ mx + b, & \text{if } x > 4 \end{cases}$ 1 / 1 point

Find the values of m and b that make $f(x)$ differentiable.

- (•) $m = 8, b = -16$
- ( ) $m = 8, b = -28$
- ( ) $m = 4, b = -12$
- ( ) $m = 4, b = 0$

Nice work

In order for f to be differentiable, f must be continuous.

We have $f(4) = 4^2$.

Thus we need

$$\lim_{x \to 4^-} f(x) = \lim_{x \to 4^+} f(x)$$
$$\lim_{x \to 4^-} x^2 = \lim_{x \to 4^+} mx + b$$
$$16 = 4m + b$$

Now we must find another condition on the constants so that f is differentiable at 4.

For $x < 4$,

$$f'(x) = \lim_{h \to 0} \frac{(x + h)^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x$$

This means f is differentiable for all $x < 4$.

For $x > 4$,

$$f'(x) = \lim_{h \to 0} \frac{m(x + h) + b - (mx + b)}{h} = \lim_{h \to 0} \frac{mh}{h} = m$$

This means f is differentiable for all $x > 4$.

We need to determine a value for m such that f will be differentiable at 4. This means

$$\lim_{x \to 4^-} f'(x) = \lim_{x \to 4^+} f'(x)$$
$$\lim_{x \to 4^-} 2x = \lim_{x \to 4^+} m$$
$$8 = m$$

Since $16 = 4m + b$, we have $16 = 4 \cdot 8 + b$ so $b = -16$.

9. Find the derivative of the function using the definition of derivative. 1 / 1 point

$g(t) = \frac{1}{\sqrt{t}}$

$-\frac{1}{2t^{\frac{3}{2}}}$

t^(-3/2)/2

Nice work

$$g'(t) = \lim_{h \to 0} \frac{g(t + h) - g(t)}{h} = \lim_{h \to 0} \frac{\frac{1}{\sqrt{t+h}} - \frac{1}{\sqrt{t}}}{h} = \lim_{h \to 0} \frac{\sqrt{t} - \sqrt{t+h}}{h\sqrt{t}\sqrt{t+h}}$$
$$= \lim_{h \to 0} \frac{\sqrt{t} - \sqrt{t+h}}{h\sqrt{t}\sqrt{t+h}} \cdot \frac{\sqrt{t} + \sqrt{t+h}}{\sqrt{t} + \sqrt{t+h}} = \lim_{h \to 0} \frac{t - (t+h)}{h(t\sqrt{t+h} + (t+h)\sqrt{t})}$$
$$= \lim_{h \to 0} \frac{-1}{t\sqrt{t+h} + (t+h)\sqrt{t}} = \frac{-1}{2t\sqrt{t}}$$

10. Find the derivative of the function using the definition of the derivative. 1 / 1 point

$k(x) = 1 - x^2$

Enter the expression for the derivative only, not k'(x) =.

$-2x$

-2*x

Nice work

$$k'(x) = \lim_{h \to 0} \frac{k(x + h) - k(x)}{h}$$
$$= \lim_{h \to 0} \frac{[1 - (x + h)^2] - [1 - x^2]}{h}$$
$$= \lim_{h \to 0} \frac{-(x^2 + 2xh + h^2) + x^2}{h}$$
$$= \lim_{h \to 0} \frac{-2xh - h^2}{h}$$
$$= \lim_{h \to 0} (-2x + h)$$
$$= -2x$$